repABC Origin Maps — PRIMARY

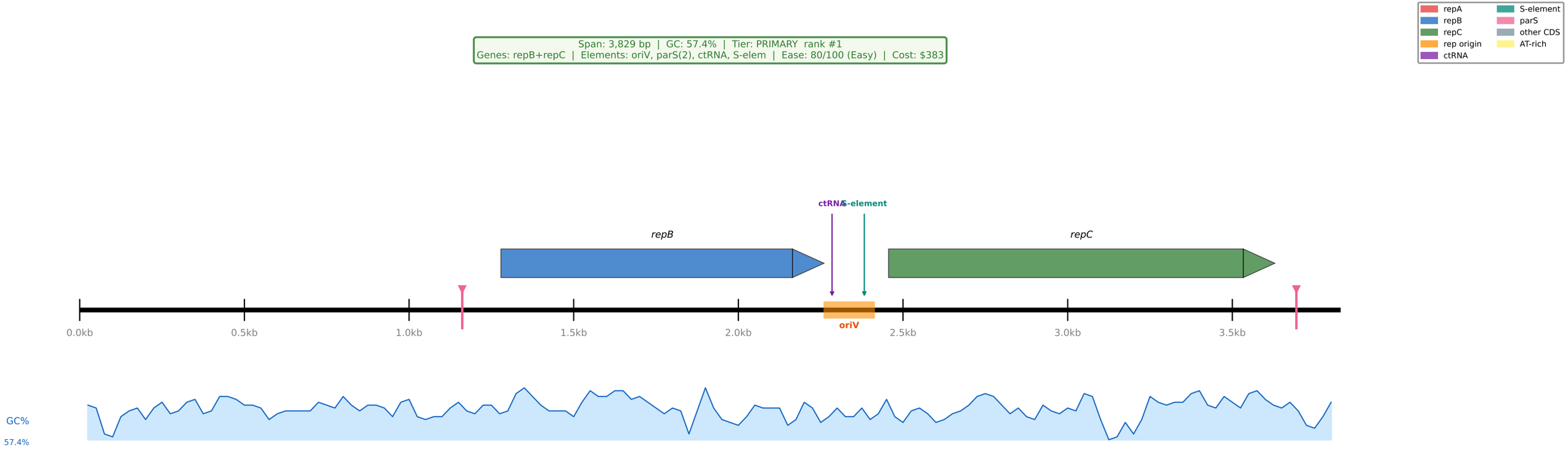
33 origins | 134,474 bp total

Synthesis ease: {'Easy': np.int64(32), 'Moderate': np.int64(1)} | Est. cost: $13,447

Rep classes: 30 | Strains: 21

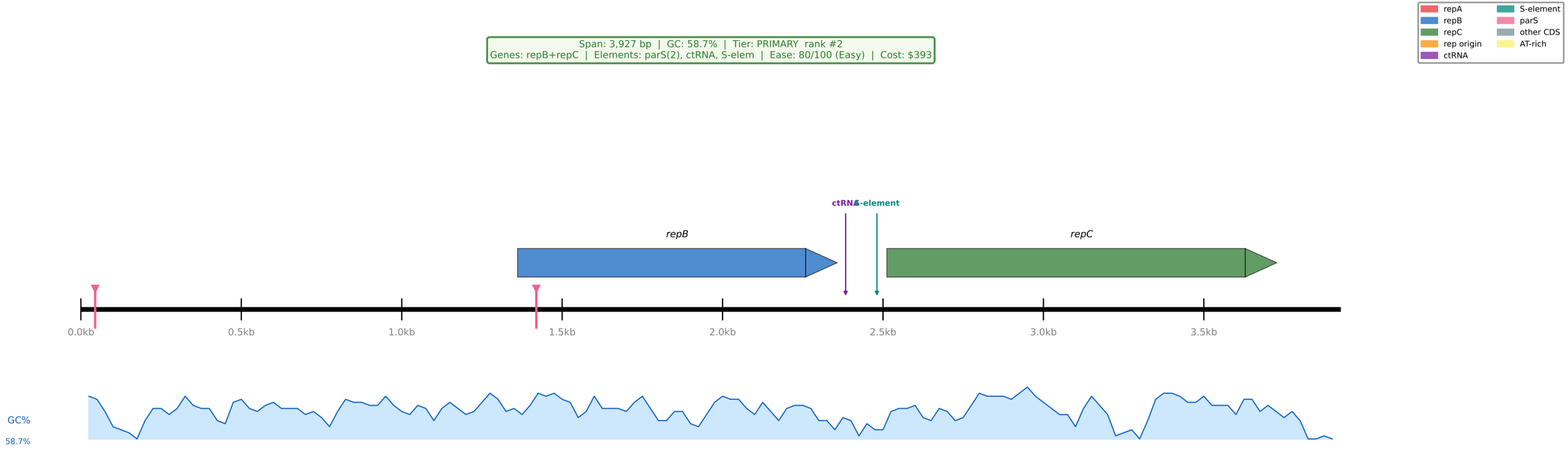
Sinorhizobium_fredii_NGR234_U00090.2 | unknown | Sinorhizobium fredii NGR234

Span: 3,829 bp | GC: 57.4% | Tier: PRIMARY rank #1
Genes: repB+repC | Elements: oriV, parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $383



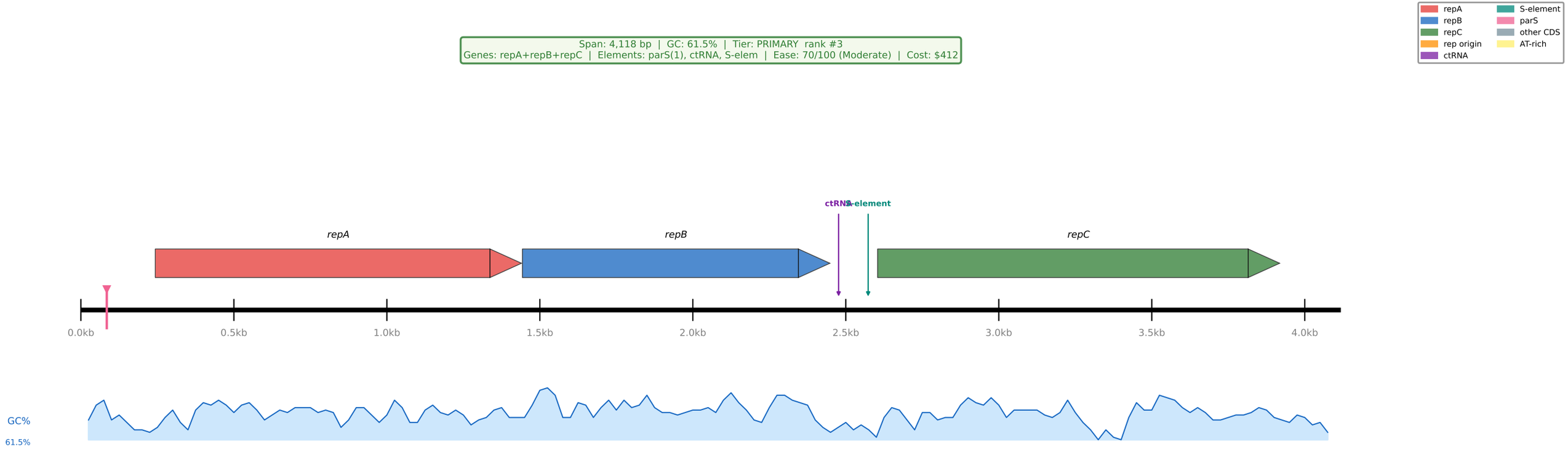
WS163_contig_4 | unknown | Agrobacterium tumefaciens

Span: 3,927 bp | GC: 58.7% | Tier: PRIMARY rank #2
Genes: repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $393



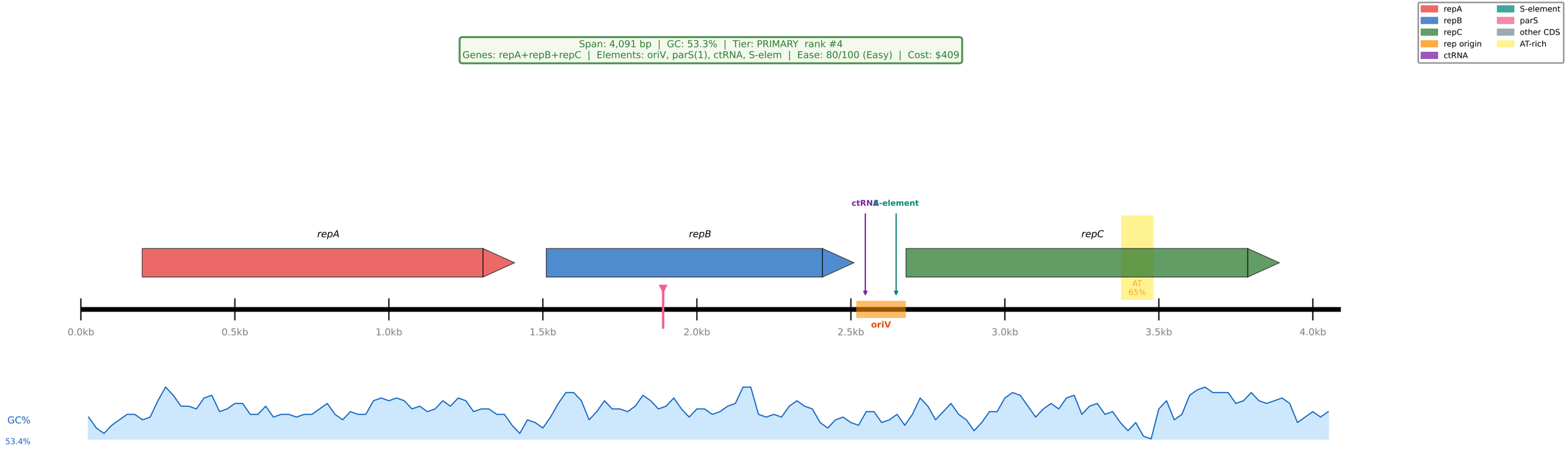
Sinorhizobium_fredii_NGR234_CP000874.1 | unknown | Sinorhizobium fredii NGR234

Span: 4,118 bp | GC: 61.5% | Tier: PRIMARY rank #3
Genes: repA+repB+repC | Elements: parS(1), ctRNA, S-elem | Ease: 70/100 (Moderate) | Cost: $412



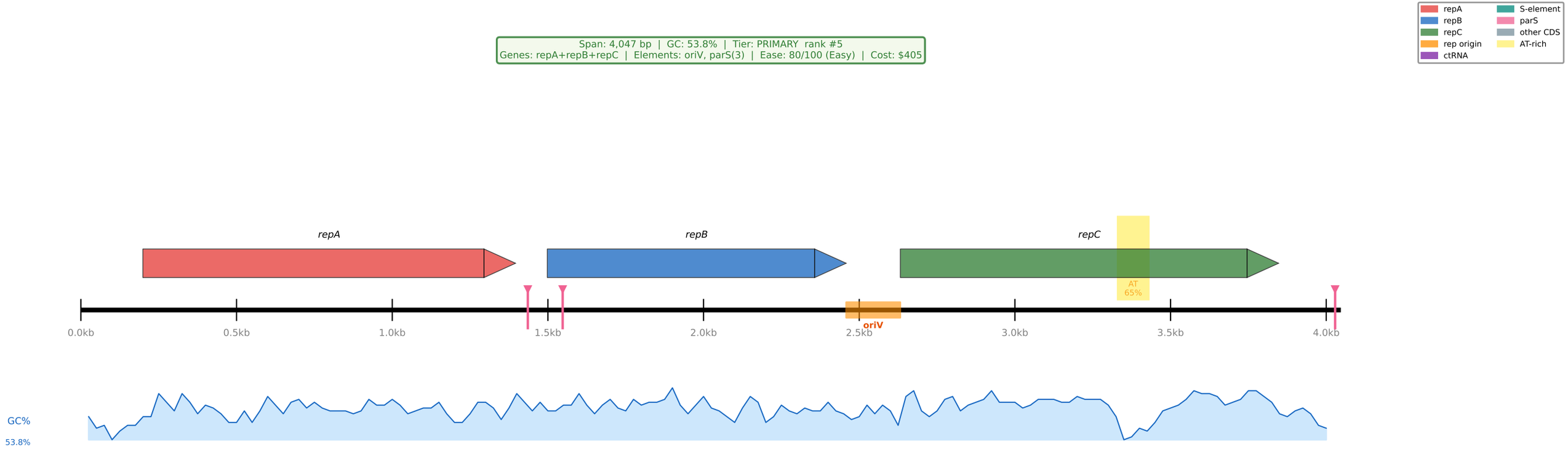
AF3.44_contig_3_pTi_Type_X | Type X | Agrobacterium tumefaciens

Span: 4,091 bp | GC: 53.3% | Tier: PRIMARY rank #4
Genes: repA+repB+repC | Elements: oriV, parS(1), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $409



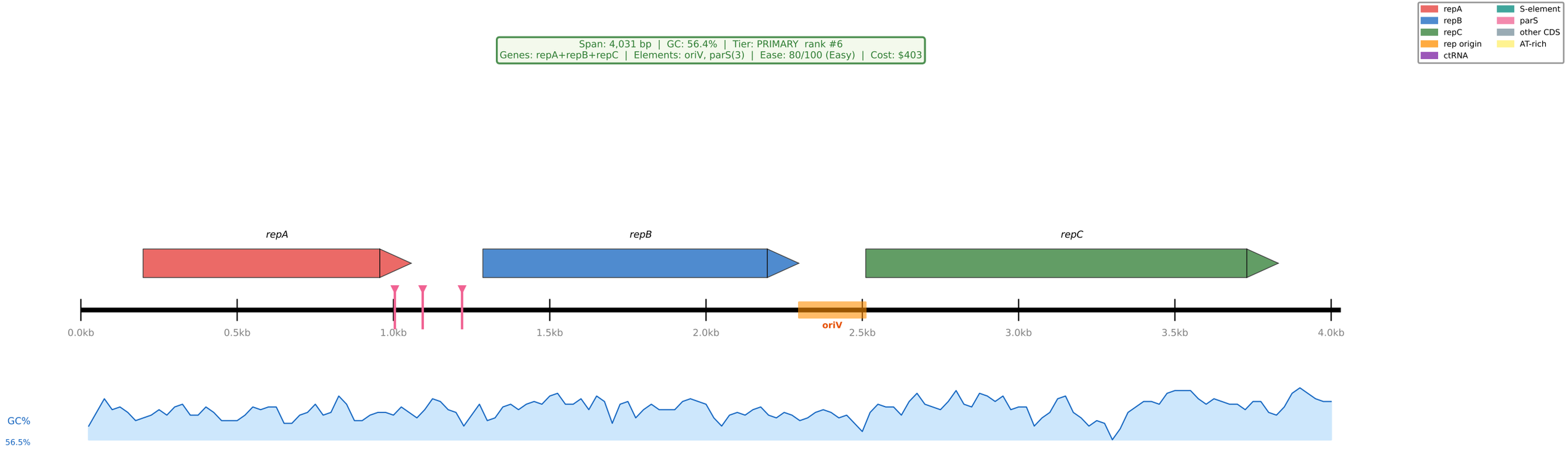
15-1187-1-2a_CP123842.1_pTi_Type_III_rep_2 | Type III rep 2 | Agrobacterium tumefaciens

Span: 4,047 bp | GC: 53.8% | Tier: PRIMARY rank #5
Genes: repA+repB+repC | Elements: oriV, parS(3) | Ease: 80/100 (Easy) | Cost: $405



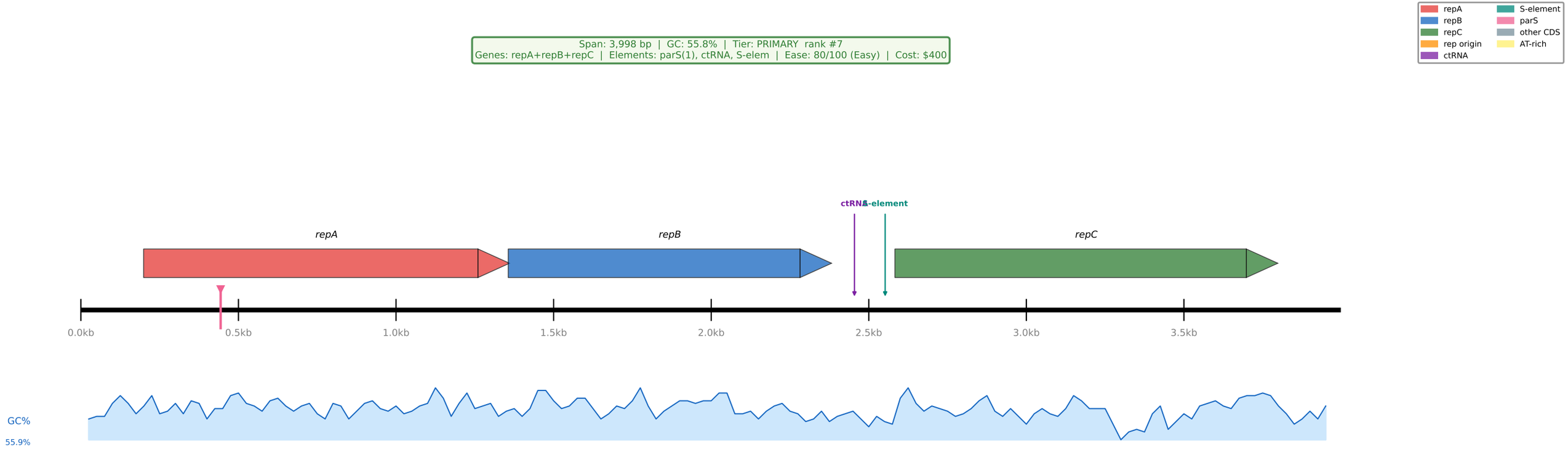
C58_AE007871.2_pTi_Type_I.a | Type I.a | Agrobacterium fabrum

Span: 4,031 bp | GC: 56.4% | Tier: PRIMARY rank #6
Genes: repA+repB+repC | Elements: oriV, parS(3) | Ease: 80/100 (Easy) | Cost: $403



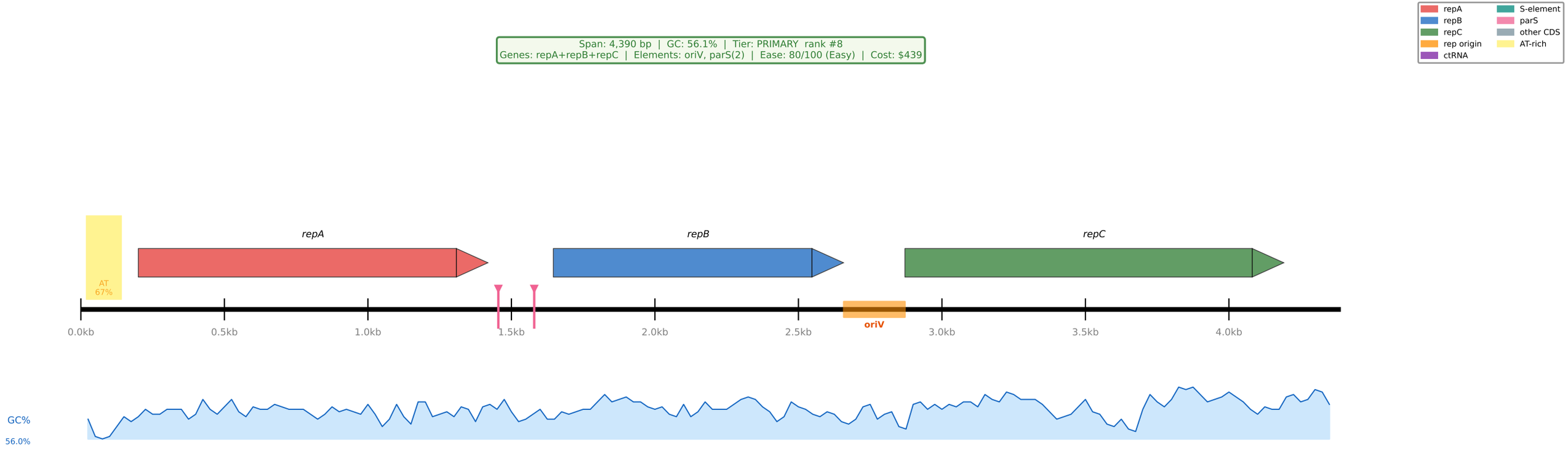
NCPPB_1641_NZ_LT009777.1_pTi_Type_XI | Type XI | Agrobacterium vitis

Span: 3,998 bp | GC: 55.8% | Tier: PRIMARY rank #7
Genes: repA+repB+repC | Elements: parS(1), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $400



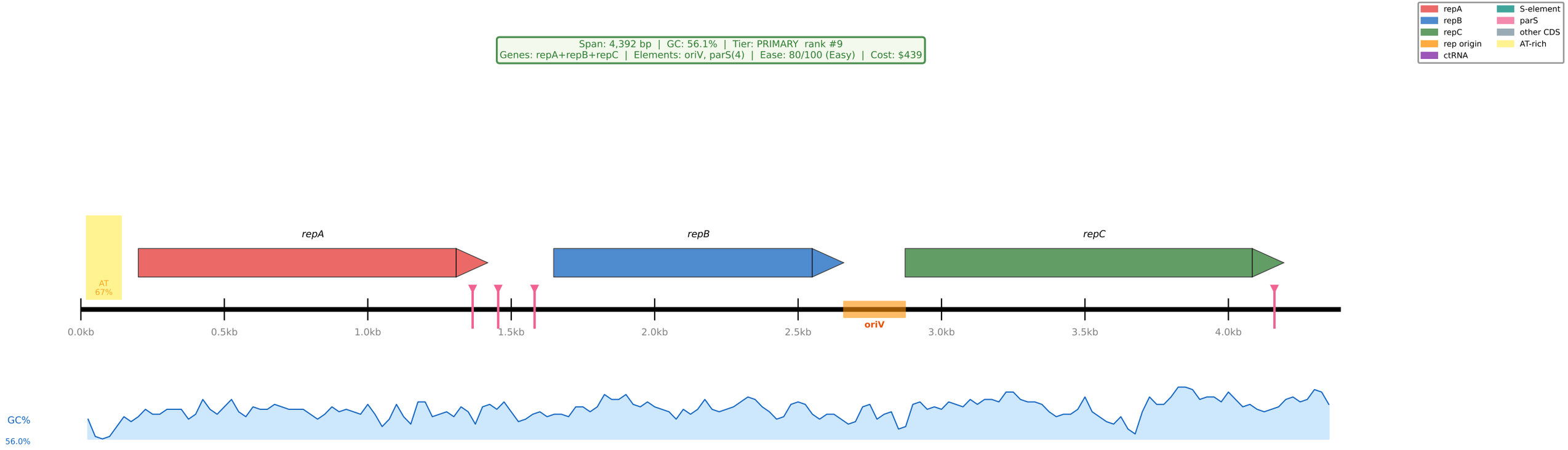
1D1524_plasmid2_pTi_Type_IV.a | Type IV.a | Agrobacterium tumefaciens

Span: 4,390 bp | GC: 56.1% | Tier: PRIMARY rank #8
Genes: repA+repB+repC | Elements: oriV, parS(2) | Ease: 80/100 (Easy) | Cost: $439



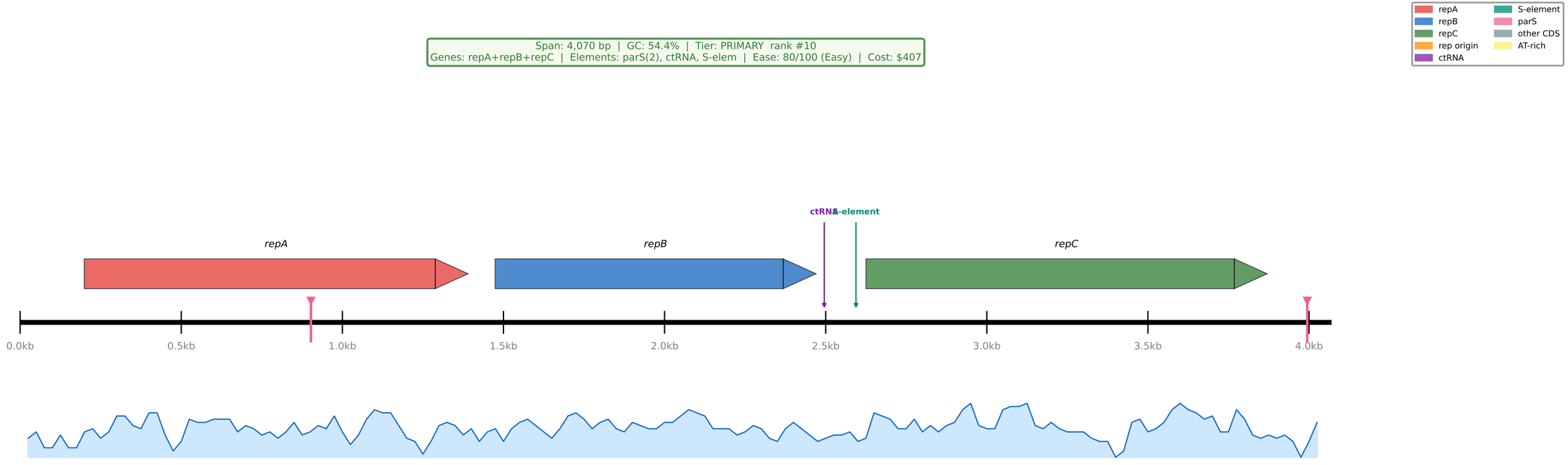
AMS008_plasmid3_pTi_Type_IV.b | Type IV.b | Agrobacterium tumefaciens

Span: 4,392 bp | GC: 56.1% | Tier: PRIMARY rank #9
Genes: repA+repB+repC | Elements: oriV, parS(4) | Ease: 80/100 (Easy) | Cost: $439



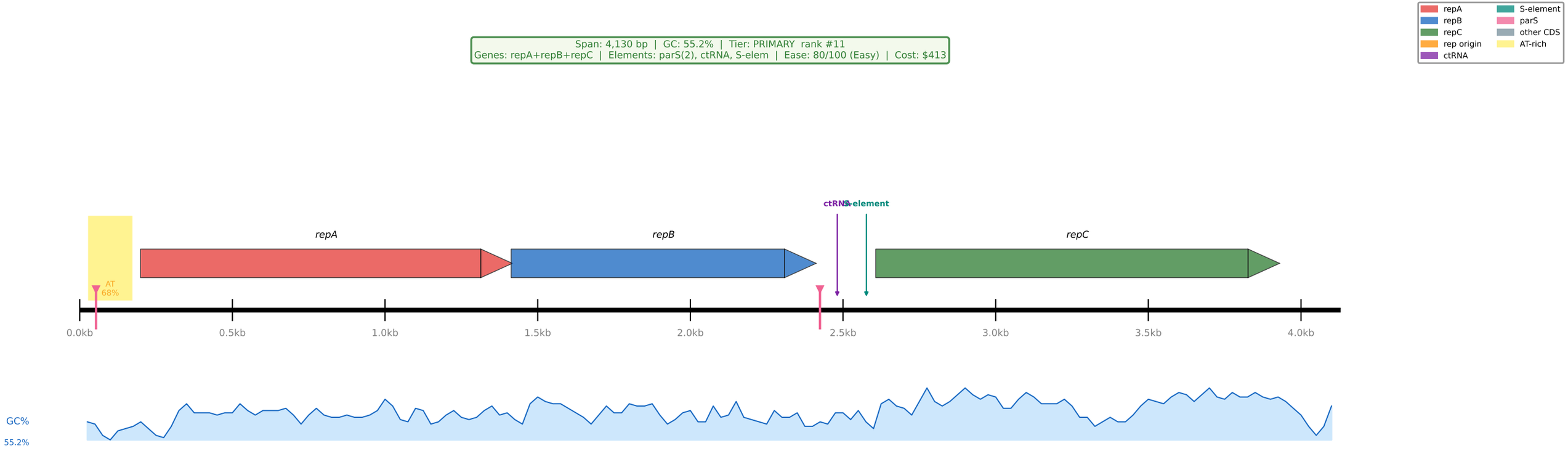
B230_85_contig_5_pTi_Type_VIII.a | Type VIII.a | Agrobacterium tumefaciens

Span: 4,070 bp | GC: 54.4% | Tier: PRIMARY rank #10
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $407



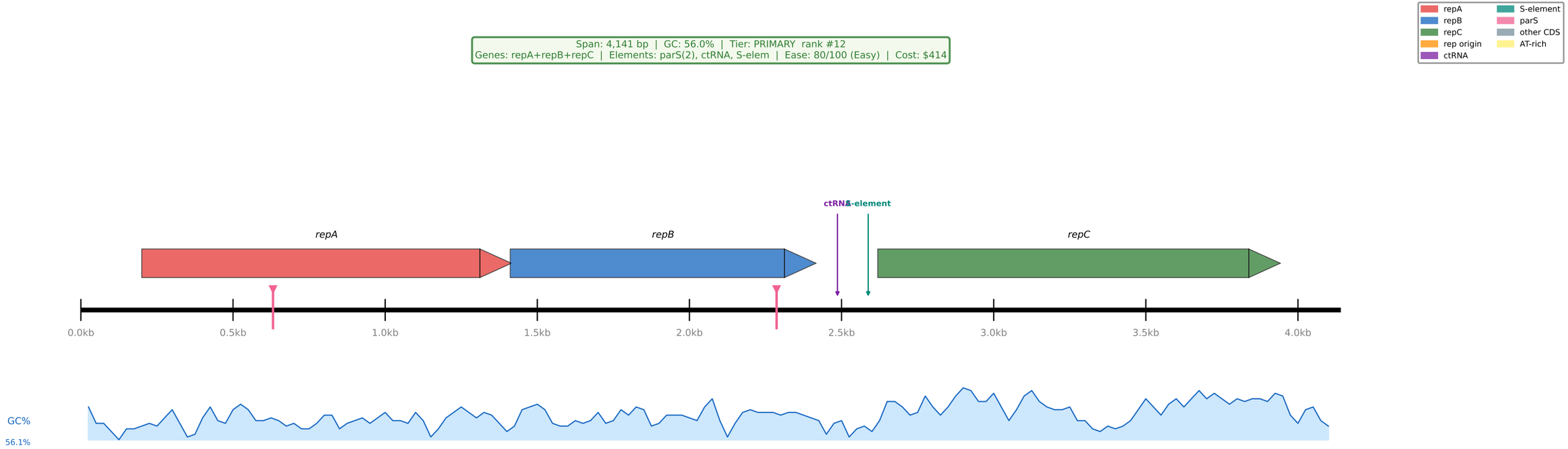
L51_94_contig_5_pTi_Type_VIII.b | Type VIII.b | Agrobacterium tumefaciens

Span: 4,130 bp | GC: 55.2% | Tier: PRIMARY rank #11
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $413



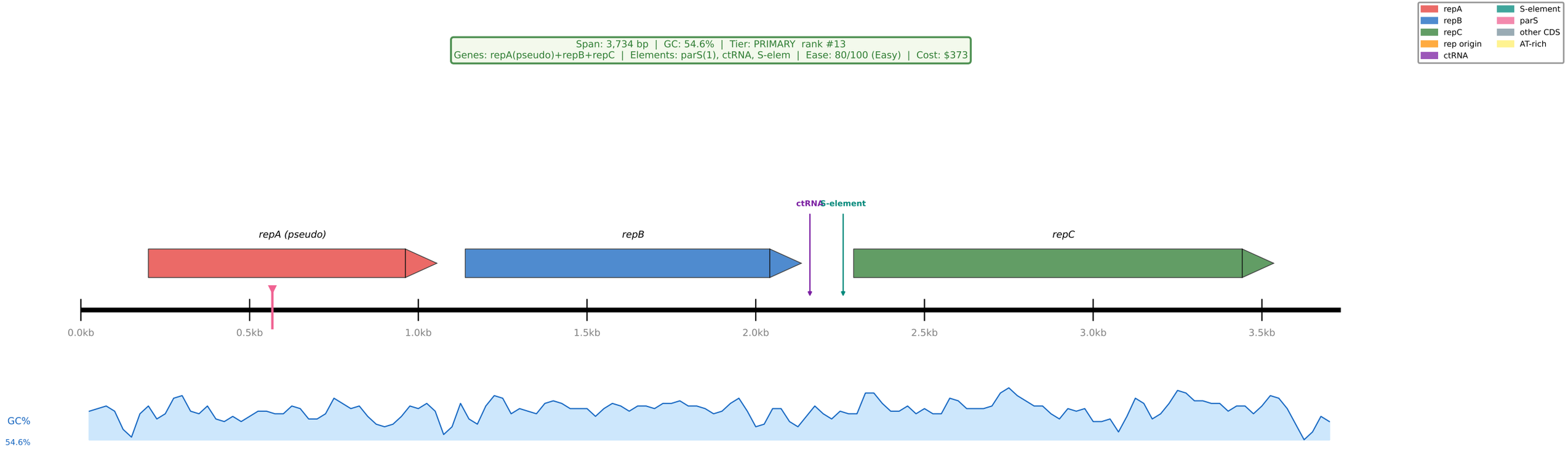
B21_90_contig_3_pTi_Type_VII | Type VII | Agrobacterium tumefaciens

Span: 4,141 bp | GC: 56.0% | Tier: PRIMARY rank #12
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $414



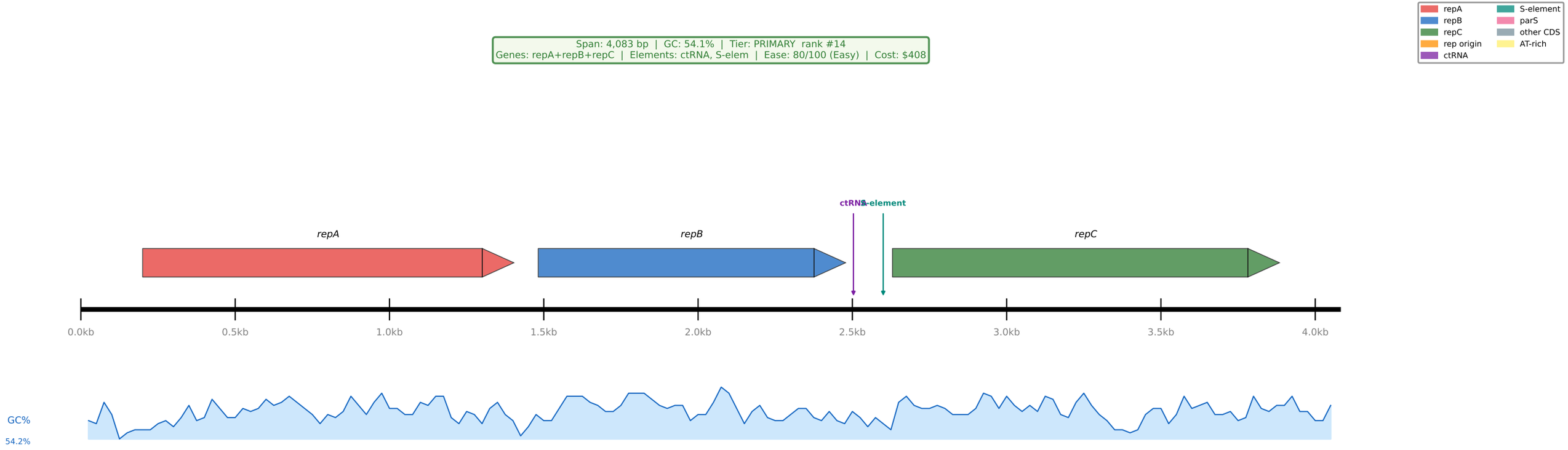
1D1514_plasmid2_pTi_Type_V_rep_1 | Type V rep 1 | Agrobacterium tumefaciens

Span: 3,734 bp | GC: 54.6% | Tier: PRIMARY rank #13
Genes: repA(pseudo)+repB+repC | Elements: parS(1), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $373



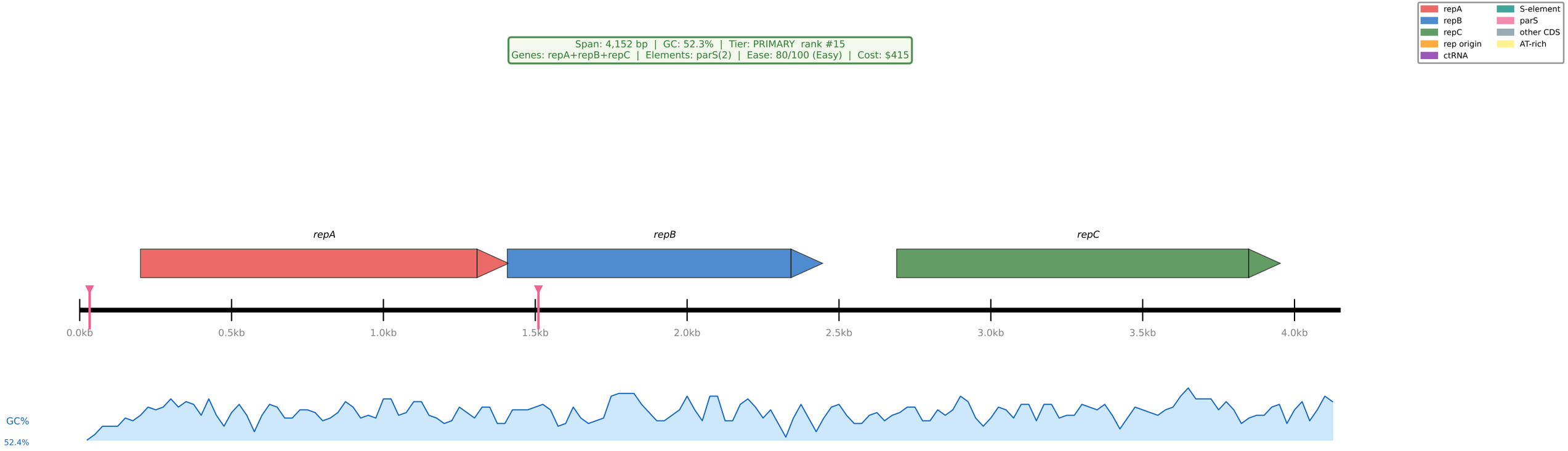
S7_73_contig_3_pTi_Type_IX | Type IX | Agrobacterium tumefaciens

Span: 4,083 bp | GC: 54.1% | Tier: PRIMARY rank #14
Genes: repA+repB+repC | Elements: ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $408



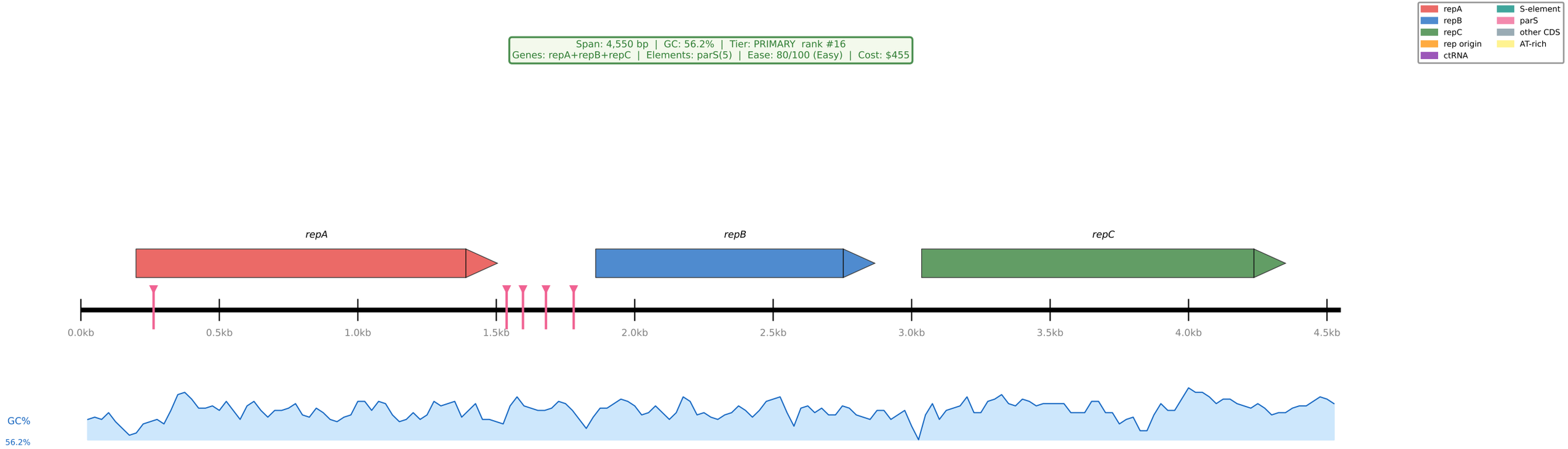
1D1514_plasmid2_pTi_Type_V_rep_2 | Type V rep 2 | Agrobacterium tumefaciens

Span: 4,152 bp | GC: 52.3% | Tier: PRIMARY rank #15
Genes: repA+repB+repC | Elements: parS(2) | Ease: 80/100 (Easy) | Cost: $415



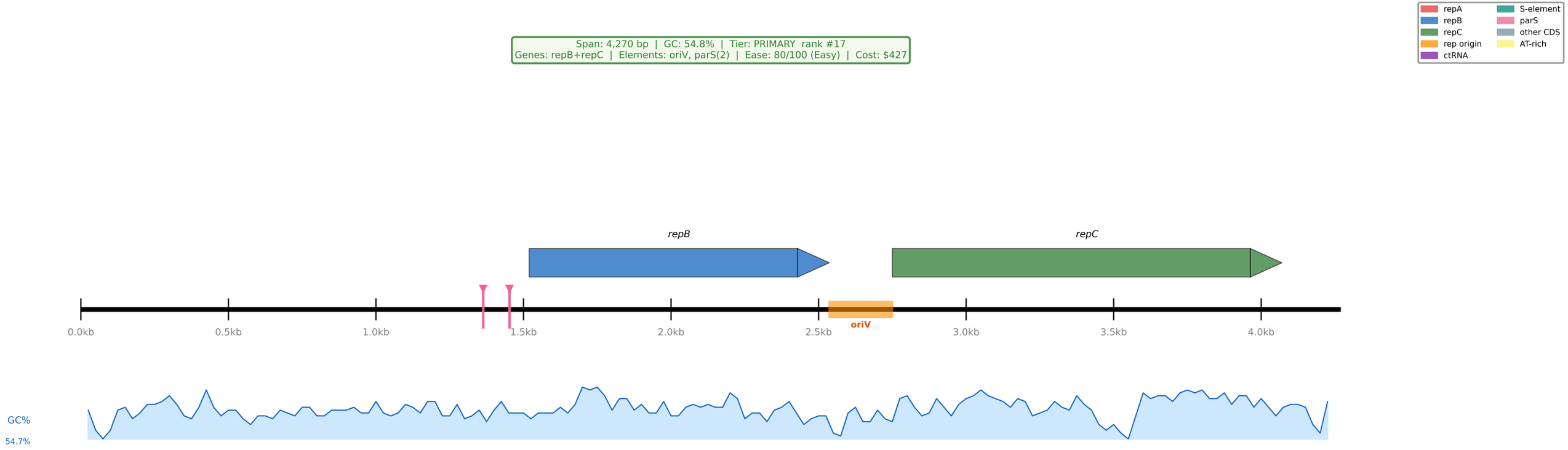
CG474_CG474_3_pTi_Type_I.c | Type I.c | Agrobacterium tumefaciens

Span: 4,550 bp | GC: 56.2% | Tier: PRIMARY rank #16
Genes: repA+repB+repC | Elements: parS(5) | Ease: 80/100 (Easy) | Cost: $455



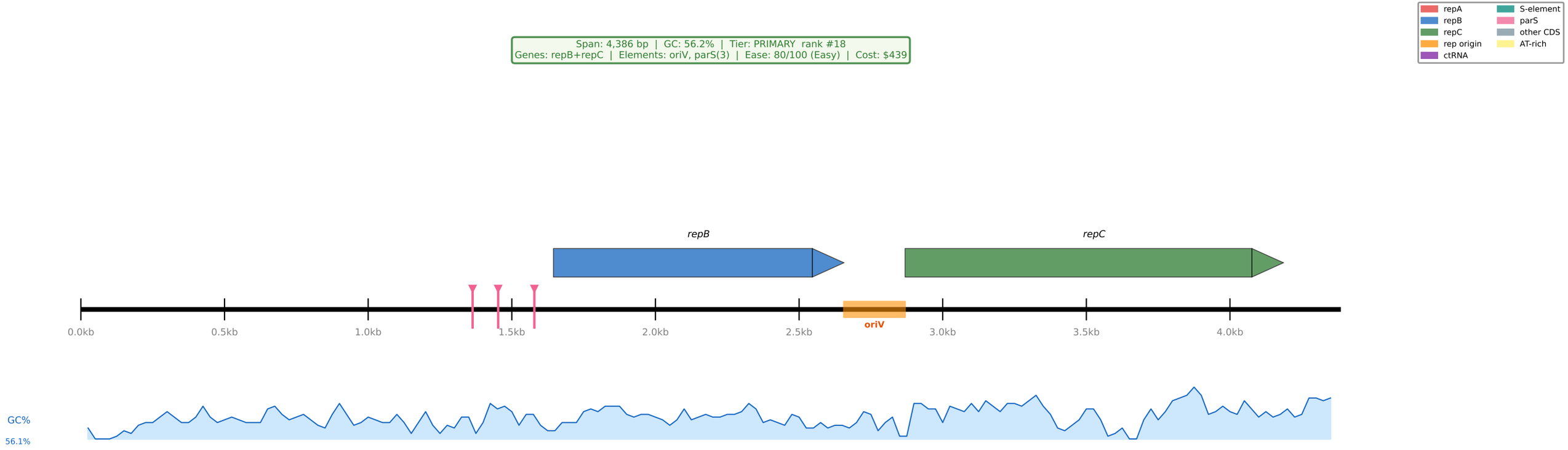
15-1187-1-2a_CP123842.1_pTi_Type_III_rep_1 | Type III rep 1 | Agrobacterium tumefaciens

Span: 4,270 bp | GC: 54.8% | Tier: PRIMARY rank #17
Genes: repB+repC | Elements: oriV, parS(2) | Ease: 80/100 (Easy) | Cost: $427



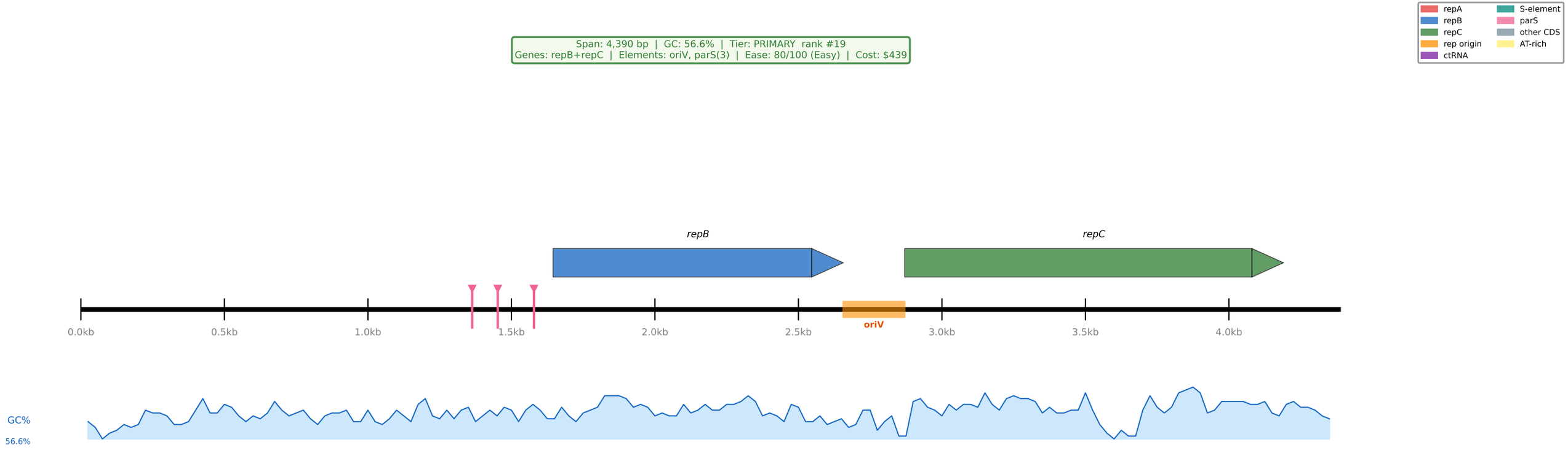
Q15_94_contig_5_pTi_Type_VI | Type VI | Agrobacterium tumefaciens

Span: 4,386 bp | GC: 56.2% | Tier: PRIMARY rank #18
Genes: repB+repC | Elements: oriV, parS(3) | Ease: 80/100 (Easy) | Cost: $439



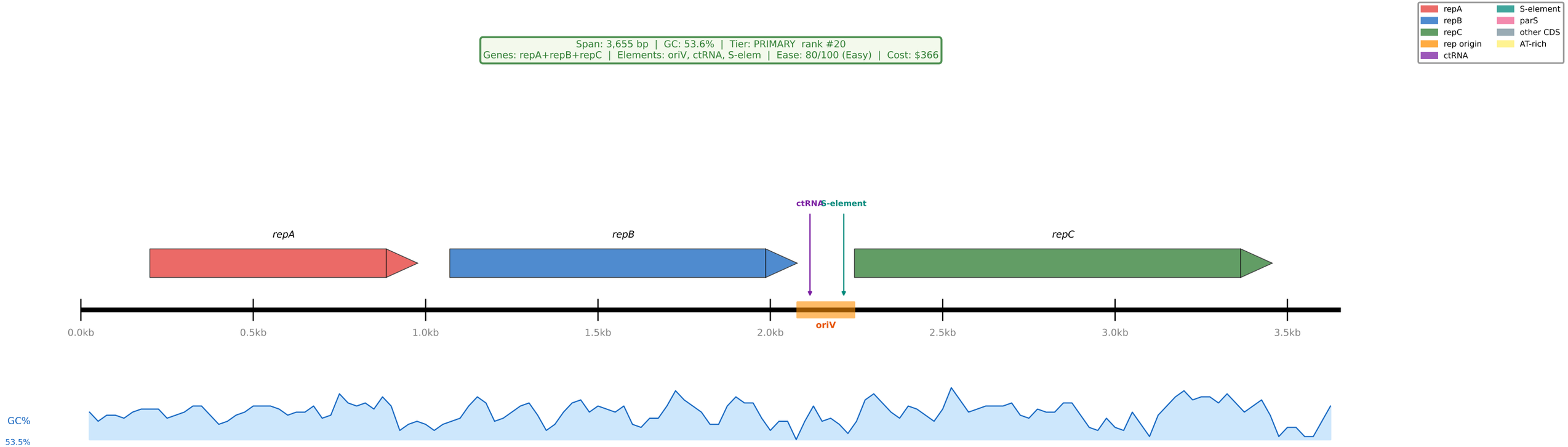
CA75_95_contig_3_pTi_Type_IV.c | Type IV.c | Agrobacterium tumefaciens

Span: 4,390 bp | GC: 56.6% | Tier: PRIMARY rank #19
Genes: repB+repC | Elements: oriV, parS(3) | Ease: 80/100 (Easy) | Cost: $439



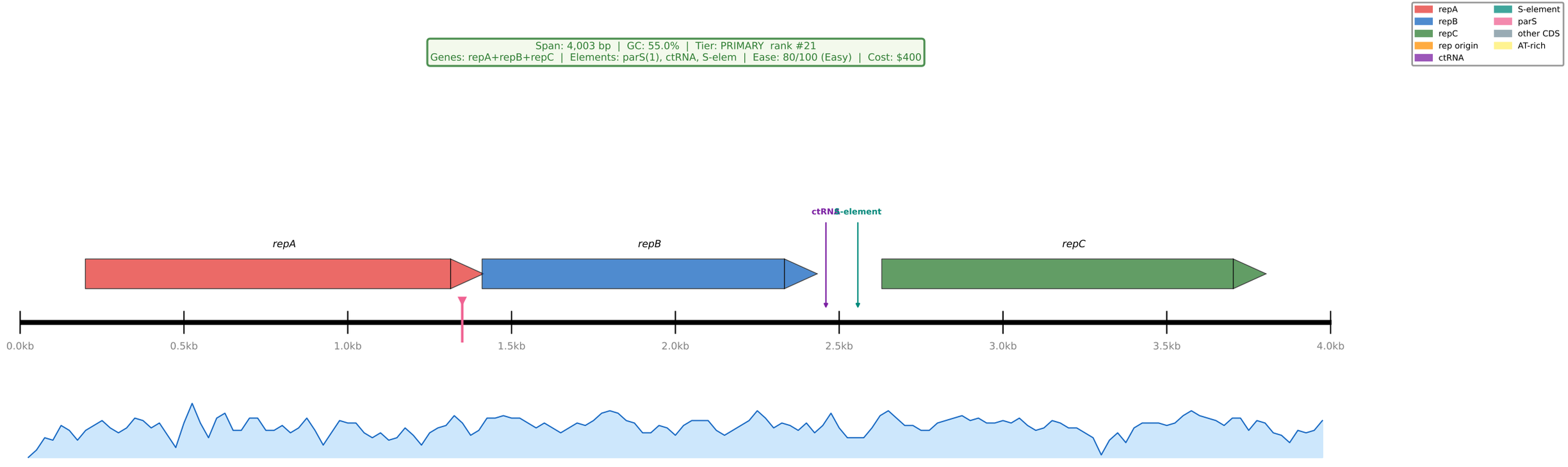
K599_CAICSX020000003.1_pRi_Type_I | Type I | Agrobacterium rhizogenes

Span: 3,655 bp | GC: 53.6% | Tier: PRIMARY rank #20
Genes: repA+repB+repC | Elements: oriV, ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $366



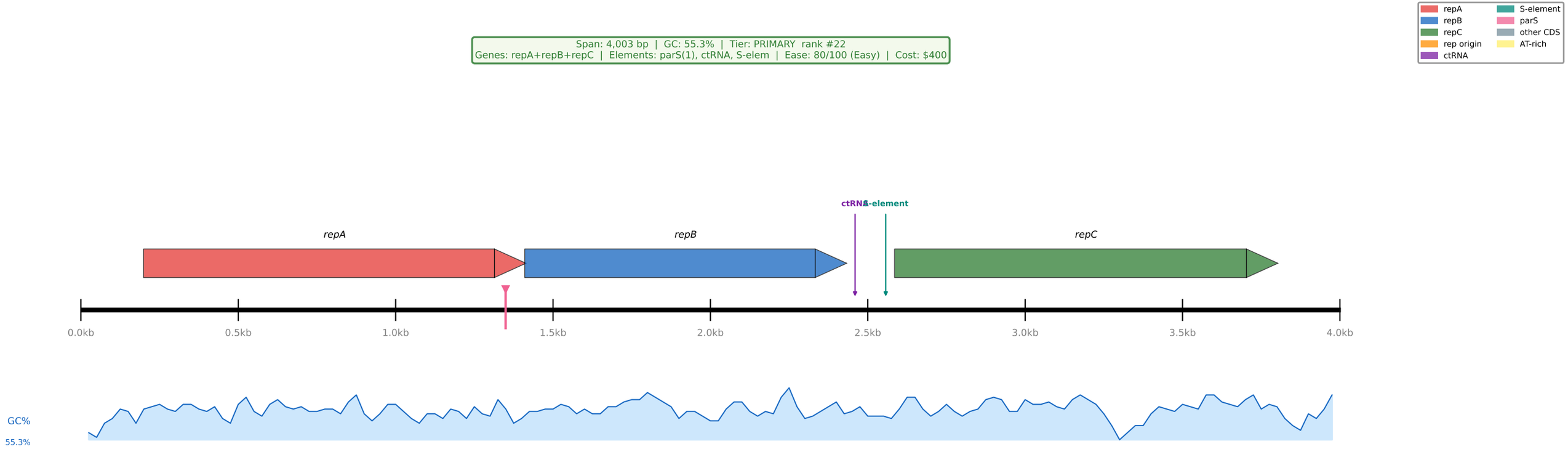
T155_95_NZ_JAAMCX010000012.1_pRi_Type_III | Type III | Agrobacterium rhizogenes

Span: 4,003 bp | GC: 55.0% | Tier: PRIMARY rank #21
Genes: repA+repB+repC | Elements: parS(1), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $400



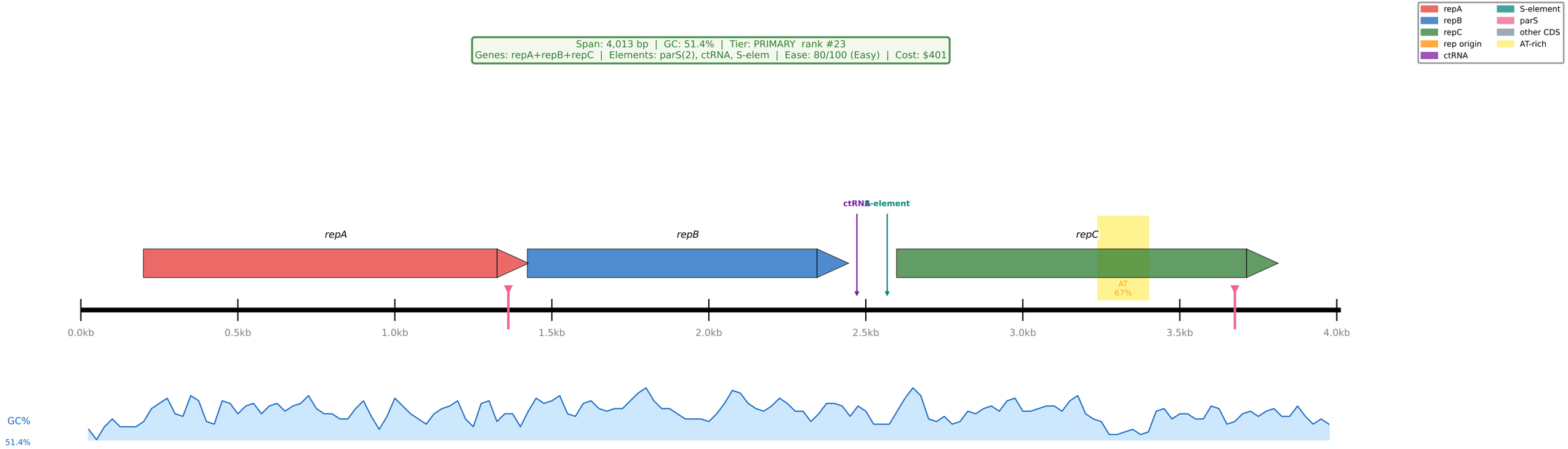
NCIB_8196_NZ_JAAMDI010000041.1_pRi_Type_II | Type II | Agrobacterium rhizogenes

Span: 4,003 bp | GC: 55.3% | Tier: PRIMARY rank #22
Genes: repA+repB+repC | Elements: parS(1), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $400



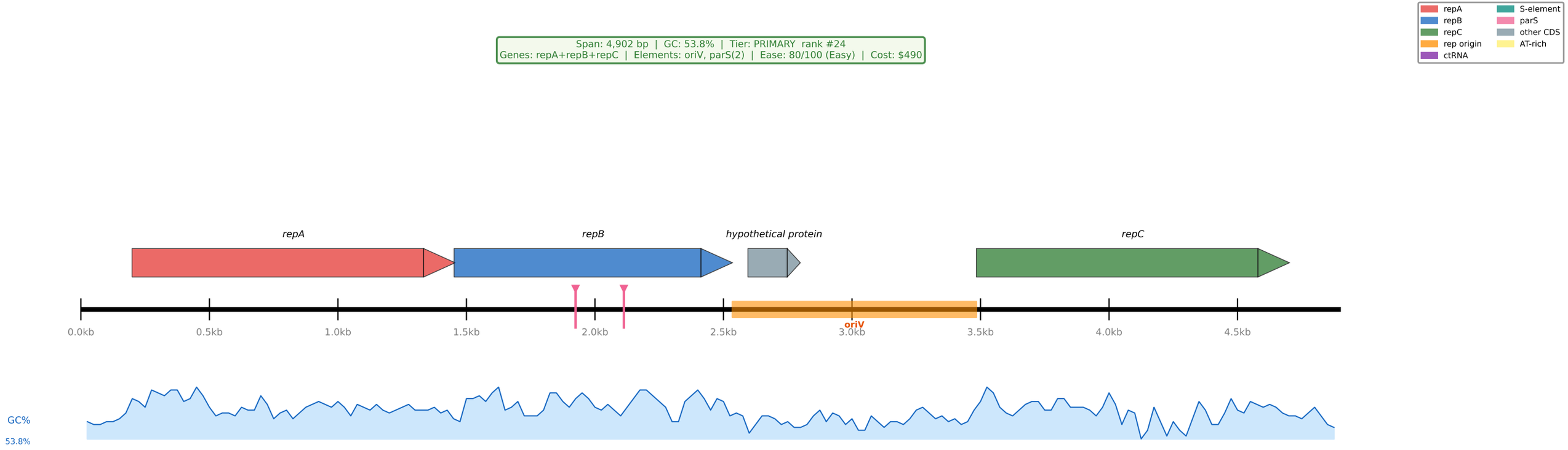
1D1524_pAt | pAt | Agrobacterium tumefaciens

Span: 4,013 bp | GC: 51.4% | Tier: PRIMARY rank #23
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $401



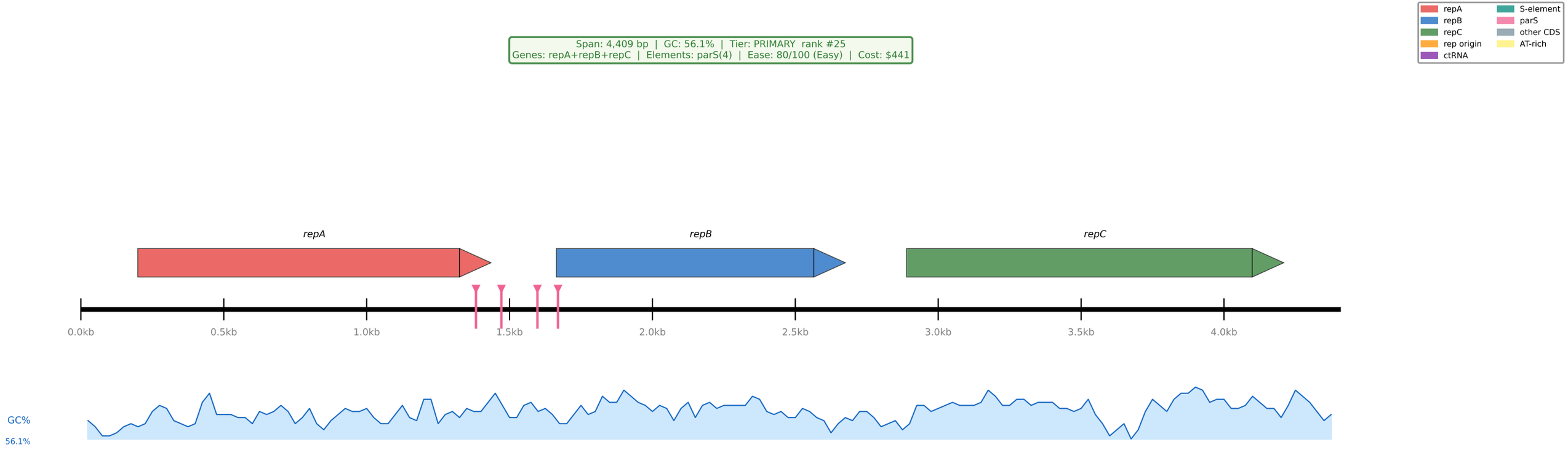
1D1524_chromid | chromid | Agrobacterium tumefaciens

Span: 4,902 bp | GC: 53.8% | Tier: PRIMARY rank #24
Genes: repA+repB+repC | Elements: oriV, parS(2) | Ease: 80/100 (Easy) | Cost: $490



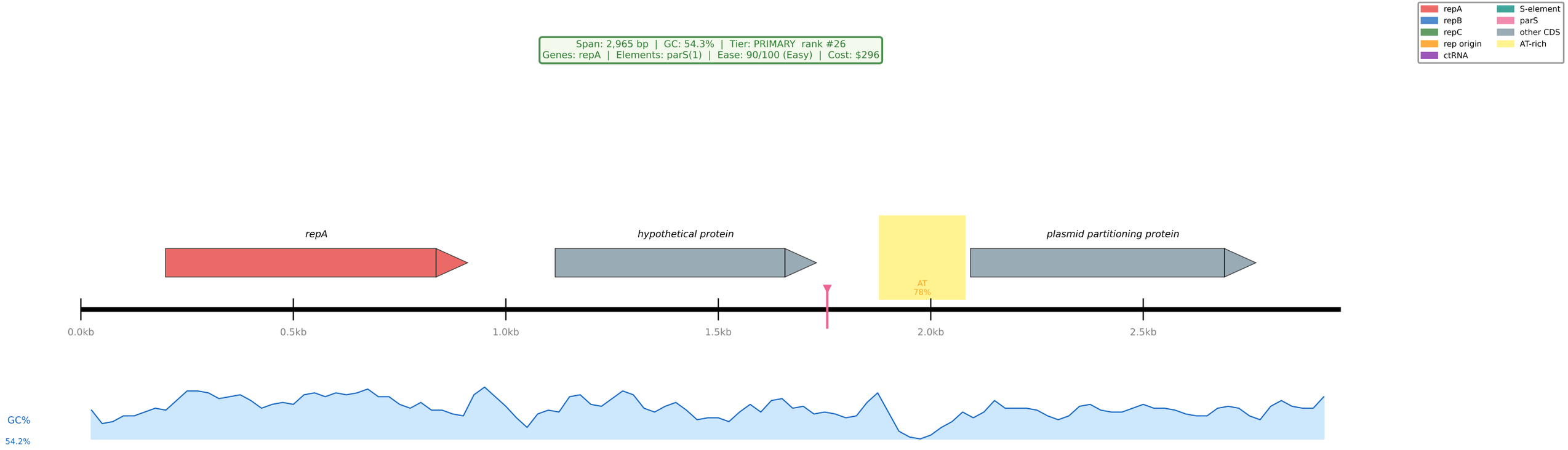
Atu_K84_pAtK84b_CP000630.1 | pAt | Agrobacterium radiobacter

Span: 4,409 bp | GC: 56.1% | Tier: PRIMARY rank #25
Genes: repA+repB+repC | Elements: parS(4) | Ease: 80/100 (Easy) | Cost: $441



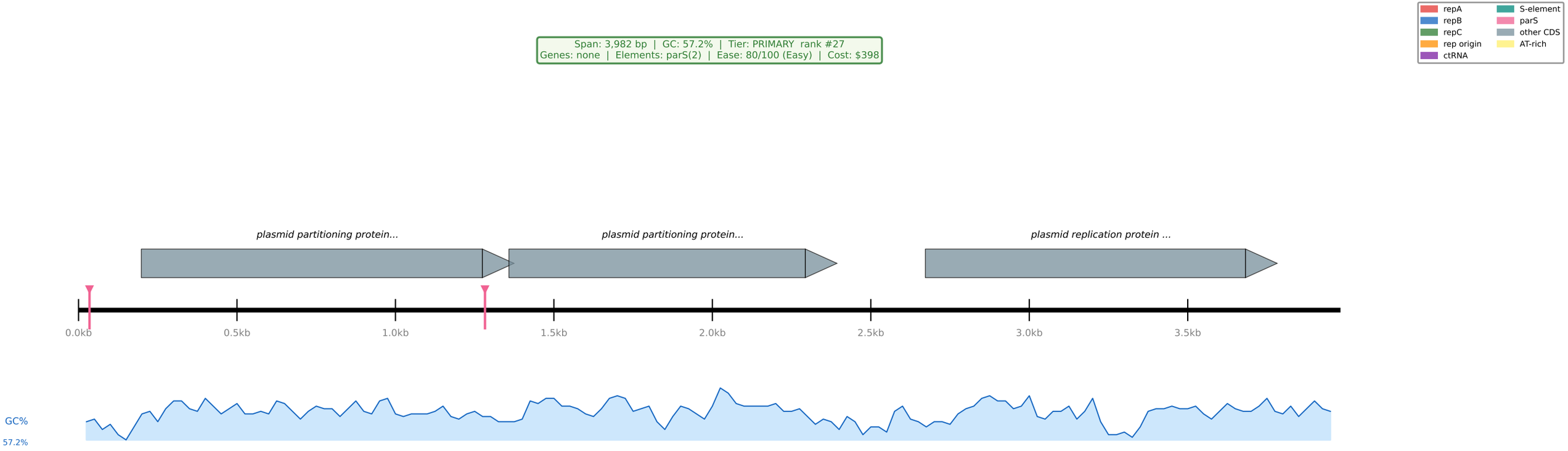
Atu_K84_pAgK84_CP000632.1 | pAg | Agrobacterium radiobacter

Span: 2,965 bp | GC: 54.3% | Tier: PRIMARY rank #26
Genes: repA | Elements: parS(1) | Ease: 90/100 (Easy) | Cost: $296



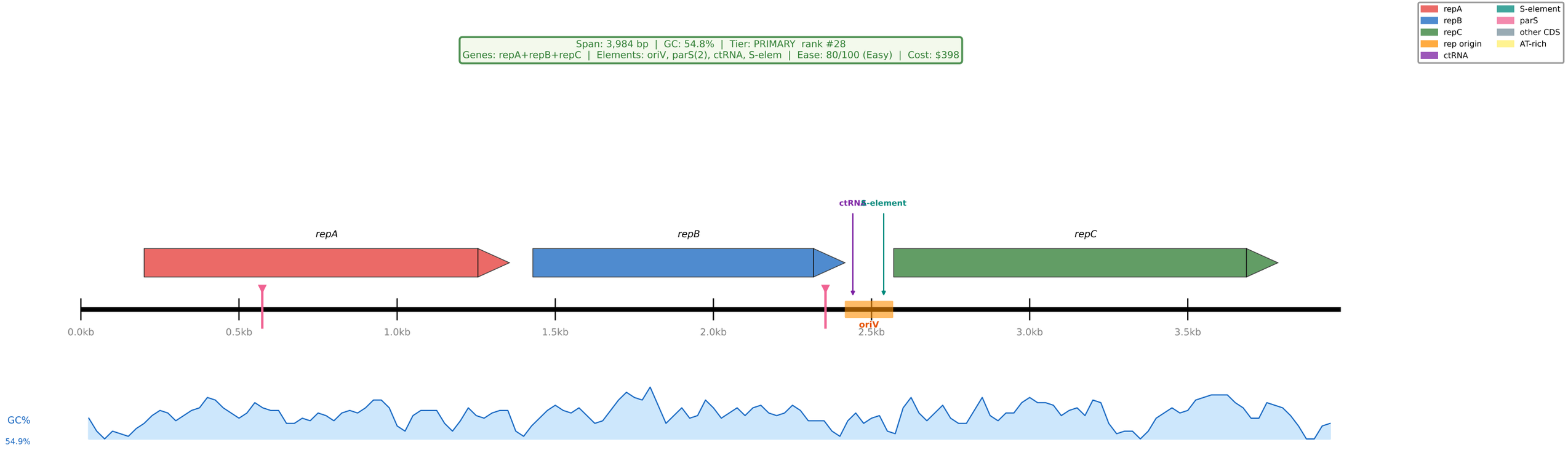
Atu_K84_chromid_CP000629.1 | chromid | Agrobacterium radiobacter

Span: 3,982 bp | GC: 57.2% | Tier: PRIMARY rank #27
Genes: none | Elements: parS(2) | Ease: 80/100 (Easy) | Cost: $398



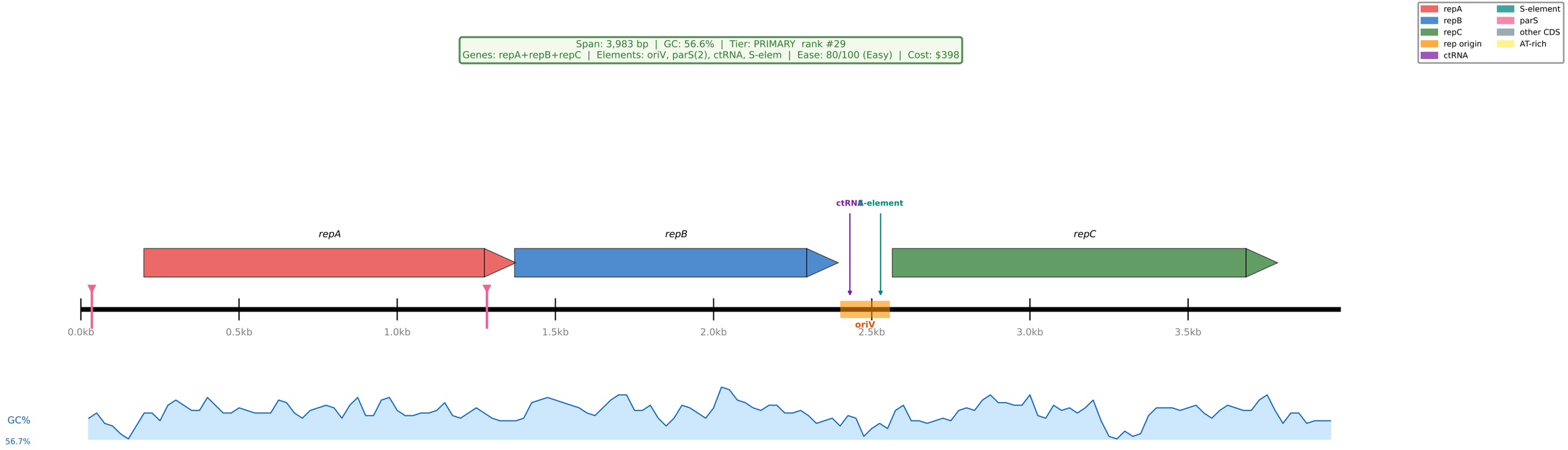
1D1514_plasmid1 | unknown | Agrobacterium tumefaciens

Span: 3,984 bp | GC: 54.8% | Tier: PRIMARY rank #28
Genes: repA+repB+repC | Elements: oriV, parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $398



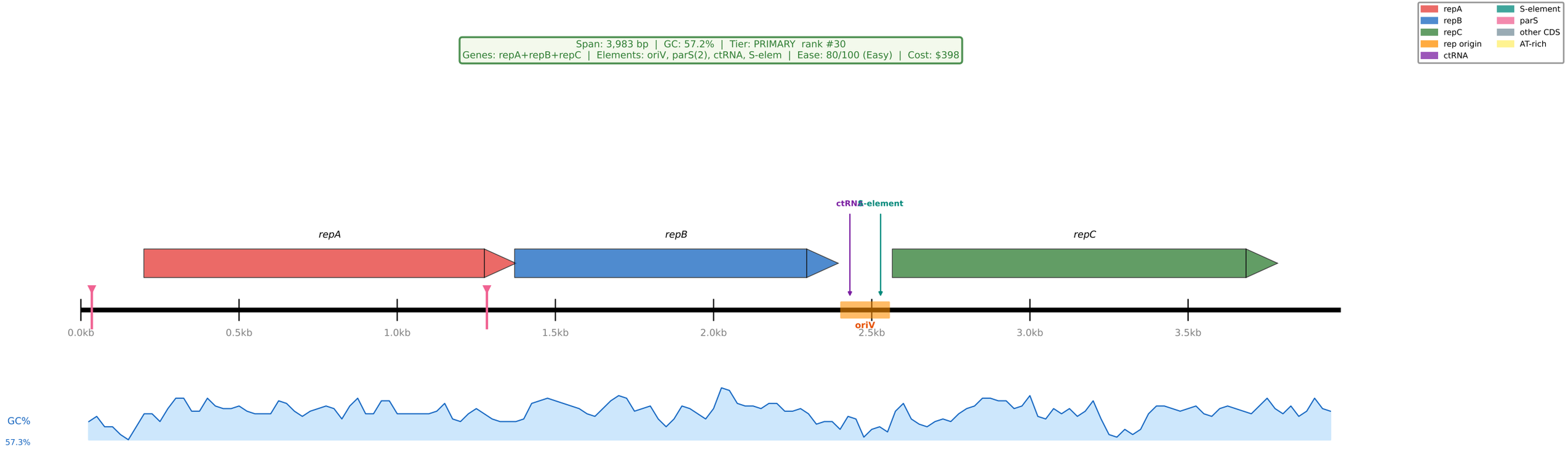
NCIB_8196_NZ_JAAMDI010000039.1 | unknown | Agrobacterium rhizogenes

Span: 3,983 bp | GC: 56.6% | Tier: PRIMARY rank #29
Genes: repA+repB+repC | Elements: oriV, parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $398



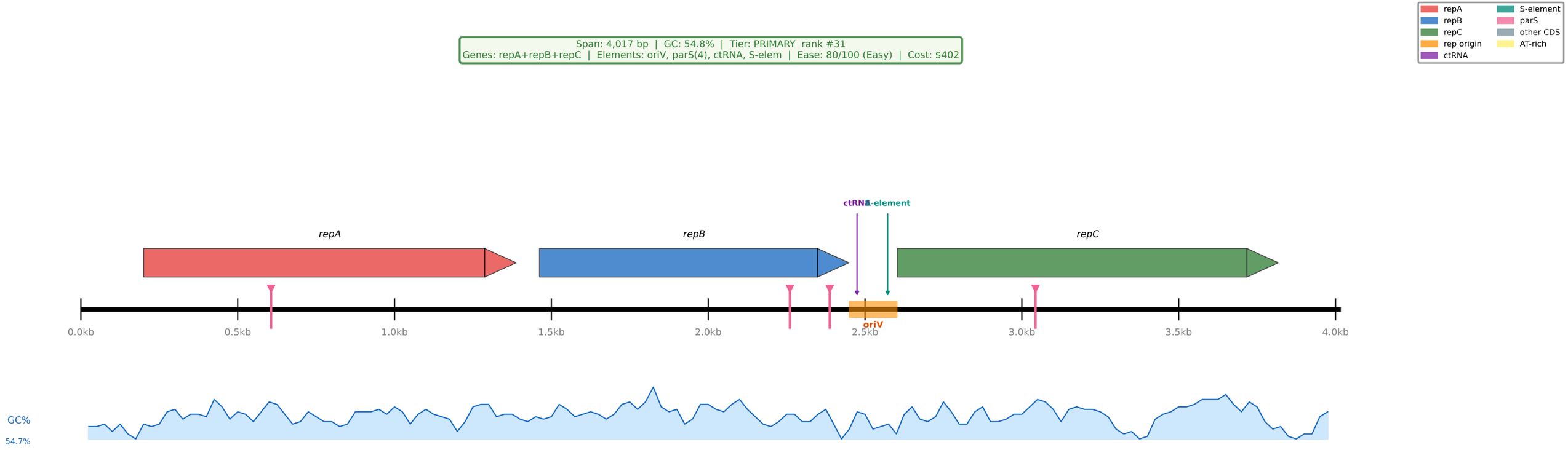
CA75_95_contig_2 | unknown | Agrobacterium tumefaciens

Span: 3,983 bp | GC: 57.2% | Tier: PRIMARY rank #30
Genes: repA+repB+repC | Elements: oriV, parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $398



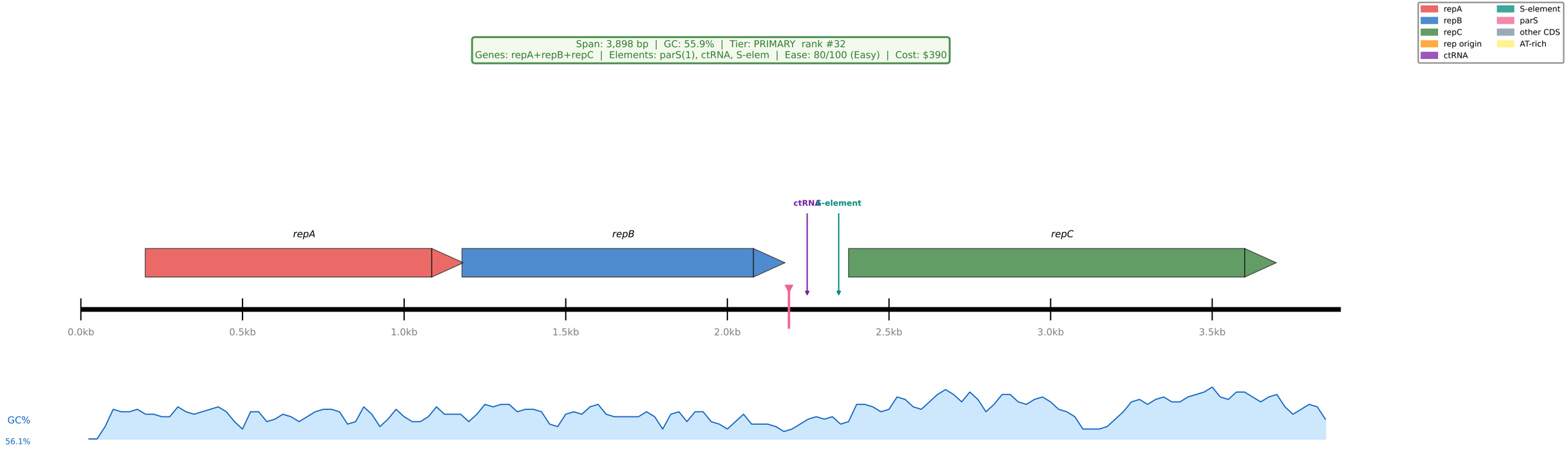
Q15_94_contig_6 | unknown | Agrobacterium tumefaciens

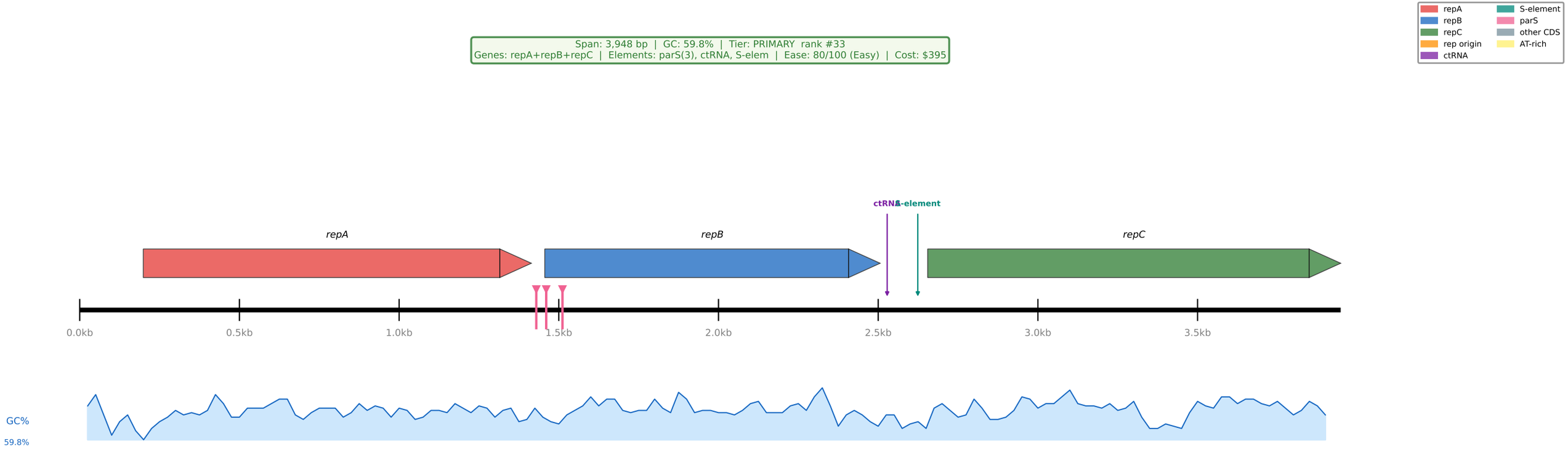
Span: 4,017 bp | GC: 54.8% | Tier: PRIMARY rank #31
Genes: repA+repB+repC | Elements: oriV, parS(4), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $402



B230_85_contig_6 | unknown | Agrobacterium tumefaciens

Span: 3,898 bp | GC: 55.9% | Tier: PRIMARY rank #32
Genes: repA+repB+repC | Elements: parS(1), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $390



Rhizobium_etli_CFN_42_CP000135.1 | unknown | Rhizobium etli CFN 42

Span: 3,948 bp | GC: 59.8% | Tier: PRIMARY rank #33
Genes: repA+repB+repC | Elements: parS(3), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $395